

ECON3500 Problem set 4

Welcome

Chapter 8 and 9 problems! Enjoy!

See the exercises below, or you can download them as a pdf.

What do I submit?

- Your written up answers to exercise questions. If you work on a piece of paper, please scan using some sort of phone software (like Microsoft Lens or Adobe Scan) rather than just taking a picture.
- A do-file that runs your Stata analysis (for question 7).
- A log file that includes the output from running your do-file (for question 7).

Exercises

1. The following equation describes the median housing price in a community in terms of amount of pollution (*nox* for nitrous oxide) and the average number of rooms in houses in the community (*rooms*):

$$\log(\text{price}) = \beta_0 + \beta_1 \log(\text{nox}) + \beta_2 \text{rooms} + u$$

- a. What are the probable signs of β_1 and β_2 ? What is the interpretation of β_1 ? Explain.
- b. Why might *nox* [or more precisely, $\log(\text{nox})$] and *rooms* be negatively correlated? If this is the case, does the simple regression of $\log(\text{price})$ on $\log(\text{nox})$ produce an upward or a downward biased estimator of β_1 ?
- c. Using data, the following equations were estimated:

$$\widehat{\log(\text{price})} = 11.71 - 1.043\log(\text{nox}), n = 506, R^2 = 0.264 \quad \widehat{\log(\text{price})} = 9.23 - 0.718\log(\text{nox}) + 0.306\text{rooms}, n = 506, R^2 = 0.514$$

Is the relationship between the simple and multiple regression estimates of the elasticity of *price* with respect to *nox* what you would have predicted, given your answer in part (b)? Does this mean that 0.718 is definitely closer to the true elasticity than 1.043?

2. Read the box “The Return to Education and the Gender Gap” in Section 8.3.
 - a. Consider a man with 16 years of education and 2 years of experience. Use the results from column (4) of Table 8.1 and the method in Key Concept 8.1 to estimate the expected change in the logarithm of average hourly earnings (AHE) associated with an additional year of experience.
 - b. Explain why your answer to (a) does not depend on the region he is from.
 - c. Repeat (a), assuming 10 years of experience.
3. To answer this question, refer to Table 8.3: *Nonlinear Regression Model of Test Scores* in your textbook:
 - a. A researcher suspects that the effect of % Eligible for subsidized lunch has a nonlinear effect on test scores. In particular, he conjectures that increases in this variable from 10% to 20% have little effect on test scores but that changes from 50% to 60% have a much larger effect.
 - i. Describe a nonlinear specification that can be used to model this form of nonlinearity.
 - ii. How would you test whether the researcher’s conjecture was better than the linear specification in column (7) of Table 8.3?
 - b. A researcher suspects that the effect of income on test scores is different in districts with small classes than in districts with large classes.
 - i. Describe a nonlinear specification that can be used to model this form of nonlinearity.

4. Labor economists studying the determinants of women's earnings discovered a puzzling empirical result. Using randomly selected employed women, they regressed earnings on the women's number of children and a set of control variables (age, education, occupation, and so forth). They found that women with more children had higher wages, controlling for these other factors. Explain how sample selection might be the cause of this result. (Hint: Notice that women who do not work outside the home are missing from the sample.) [This empirical puzzle motivated James Heckman's research on sample selection that led to his 2000 Nobel Prize in Economics. See Heckman (1974)]
5. The demand for a commodity is given by $Q = \beta_0 + \beta_1 P + u$, where Q denotes quantity, P denotes price, and u denotes factors other than price that determine demand. Supply for the commodity is given by $Q = \gamma_0 + \gamma_1 P + v$, where v denotes factors other than price that determine supply. Suppose u and v both have a mean of 0, have variances σ_u^2 and σ_v^2 , and are mutually uncorrelated.
 - a. Solve the two simultaneous equations to show how Q and P depend on u and v .
 - b. Derive the means of P and Q .
 - c. Derive the variance of P , the variance of Q , and the covariance between Q and P .
6. Revisit the box "*The Return to Education and the Gender Gap*" in Section 8.3. Discuss the internal and external validity of the estimated effect of education on earnings
7. Complete Additional Empirical Exercise 8.2 using the dataset `CollegeDistance.dta`